FINALTERM EXAMINATION
SPRING 2017

EDU431- TEST DEVELOPMENT AND ASSESSMENT

18 AUGUST 2017

Q.1:

Define the term measurement? (From Topic 1)

Answer:

Measurement:

Measurement is the process by which the attributes or dimensions of some object (both physical and abstract) are quantified.

Q.2 :

What is the table of specifications?

Answer:

One of the tools used by the teachers to develop a blueprint for the test is called the table specification.

Q.3:

Write any three limitations of MCQS .(From Topic 75)

Answer:

Limitations of MCQS:

1. Take a long time to construct in order to avoid arbitrary and ambiguous questions.
2. Also require careful preparation to avoid preponderance of questions testimony recall.
3. Provide cues that do not exist in practice.

Q.4:

What is the predictive validity?

Answer:

Predictive Validity:

Predictive validity refers that how much test can predict future performance with reference to some defined standard.

Q.5:

What is the diagnostic assessment? (From Topic 4)

Diagnostic assessment:

Diagnostic assessment is the type of the assessment that determines the causes intellectual, emotional, and environmental of persistent learning difficulties.

Q.6:

Describe the Bloom's taxonomy of cognitive domain?

Answer:

Cognitive Domain :

There are six level's of the cognitive domain:

1. Knowledge
2. Comprehension
3. Application
4. Analysis

5. Synthesis
6. Evaluation

Q.7:

What is the test security? (From Topic 106)

Answer:

Test Security:

Test security is the process intended or reveal flaws in the security mechanism of an information system that protect data and maintain functionality as intended.

Q.8:

How to design rubric?(From Topic 112)

Answer:

We can design a rubric by the following some steps:

1. Identify a learning goal.
2. Choose outcomes that may be measured.
3. Develop or adapt an existing rubric.
4. Share it with students.

Q.9:

How to assemble or packaging a test?(From Topic 103)

Answer:

Assembling Test (Packaging the Test):

1. Grouping together items of similar format
2. Arranging test items from easy to hard
3. Properly spacing items
4. Keeping items and options on the same page
5. Placing illustrations near the descriptions
6. Checking the answer key
7. Deciding how students will record their answers
8. Providing space for test taker's name and date
9. Checking test directions for clarity

10.Proofreading the test

Q.10:

Write a brief note on the restricted response essay type items?

Answer:

Pose a specific problem for which student needs to recall suitable information, organize it, derive a defensible conclusion, and express it within the given limits the questions (like 400 word, 30 min. etc.).

Example:

List the similarities and differences in the process of cell division in meiosis and mitosis.

A variety of learning outcomes (Linn, 2000) can be checked by using this format of **essay question. Some of which** are;

1. Analysis of *relationship.*
2. *Compare and contrast* positions.
3. Explain *cause-effect* relationship.
4. *Organize data* and support a viewpoint.
5. *Formulate hypotheses*.
6. Point out *strengths and weaknesses.*
7. *Integrate data* from various resources

Teacher can use this type of questions under the following conditions when:

1. *Supplying information is required* instead of simple recognition.
2. *Limited numbers of content areas are needed to be tested*.

If it is feared that students may help each other unfairly during the test which is difficult to control. (Especially if the number of students is large and class teacher cannot invigilate all of

them as keenly as needed). Student is to decide length, content to be included, its organization and order of presentation min the answer.

Q.11:

How to write essay type questions? (From Topic 90)

Answer:

Following guidelines are provided by Haldane and Downing (1989) for writing essay type items when developing a test.

1. Frame questions so that the examinee's task is explicitly defined.
2. Specify the value and an approximate time limit for each question.
3. Employ a larger number of questions that require relatively short answers rather than only a few questions that require long answers.
4. Do not employ optional questions.
5. Verify a question's quality by writing a trial response to the question.
6. Prepare a tentative scoring key in advance of considering examinee responses.
7. Score all answers to one question before scoring the next question.
8. Make prior decisions regarding treatment of such factors as spelling and punctuation.
9. Evaluate essay responses anonymously.

Q.12:

Explain Criterion and predictive Validity.

Answer:

Criterion validity refers how much test can measure some performance with reference to some defined standard.

For example:

The student is expected to be able to speak fluently after completing a course. So, the test is valid if it can measure spoken ability.

Predictive validity refers that how much test can predict future performance with reference to some defined standard.

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Q.1:

Explain the Taxonomy levels of Cognitive.

Answer:

Bloom's Taxonomy of Cognitive Development

Bloom identified six levels within the cognitive domain, from the simple recall or recognition of facts, as the lowest level, through increasingly more complex and abstract mental levels, to the highest order which is classified as evaluation. A description of the six levels as well as verb examples that represent intellectual activity are listed here.



Knowledge is defined as remembering of previously learned material. This may involve the recall of a wide range of material, from specific facts to complete theories, but all that is required is the bringing to mind of the appropriate information. Knowledge represents the lowest level of learning outcomes in the cognitive domain.

Verbs: arrange, define, duplicate, label, list, memorize, name, order, recognize, relate, recall, repeat, reproduce state.

Comprehension is defined as the ability to grasp the meaning of material. This may be shown by translating material from one form to another (words to numbers), by interpreting material (explaining or summarizing), and by estimating furture trends (predicting consequences or effects). These learning outcomes go one step beyond the simple remembering of material, and represent the lowest level of understanding.

Verbs: classify, describe, discuss, explain, express, identify, indicate, locate, recognize, report, restate, review, select, translate.

Application refers to the ability to use learned material in new and concrete situations. This may include the application of such things as rules, methods, concepts, principles, laws, and theories. Learning outcomes in this area require a higher level of understanding than those under comprehension.

Verbs: apply, choose, demonstrate, dramatize, employ, illustrate, interpret, operate, practice, schedule, sketch, solve, use, write.

Analysis refers to the ability to break down material into its component parts so that its organizational structure may be understood. This may include the identification of the parts, analysis of the relationships between parts, and recognition of the organizational principles involved. Learning outcomes here represent a higher intellectual level than comprehension and application becasue they require an understanding of both the content and the structural form of the material.

Verbs: analyze, appraise, calculate, categorize, compare, contrast, criticize, differentiate, discriminate, distinguish, examine, experiment, question, test.

Synthesis refers to the ability to put parts together to form a new whole. This may involve the production of a unique communication (theme or speech), a plan of operations (research proposal), or a set of abstract relations (scheme for classifying information). Learning outcomes in this area stress creative behaviors, with major emphasis on the formulation of new patterns or structures.

Verbs: arrange, assemble, collect, compose, construct, create, design, develop, formulate, manage, organize, plan, prepare, propose, set up, write.

Evaluation is concerned with the ability to judge the value of material (statement, novel, poem, research report) for a given purpose. The judgements are to be based on definite criteria. These may be internal criteria (organization) or external criteria (relevance to the purpose) and the student may determine the criteria or be given them. Learning outcomes in this area are highest in the cognitive hierarchy because they contain elements of all the other categories, plus conscious value judgements based on clearly defined criteria.

Verbs: appraise, argue, assess, attach, choose compare, defend estimate, judge, predict, rate, core, select, support, value, evaluate.

Reference*: Major categories in the cognitive domain of the taxonomy of educational objectives* (Bloom, 1956).

Task Analysis

Q.2:

What is the formula for the find out the item discrimination?(From Topic 97)

Answer:

ID is = (Upper Group % Correct) – (Lower Group % Correct)

Q.3:

What are the guidelines for the essay type questions? (Repeat)

Answer:

Following guidelines are provided by Haldane and Downing (1989) for writing essay type items when developing a test.

10. Frame questions so that the examinee's task is explicitly defined.
11. Specify the value and an approximate time limit for each question.
12. Employ a larger number of questions that require relatively short answers rather than only a few questions that require long answers.
13. Do not employ optional questions.
14. Verify a question's quality by writing a trial response to the question.
15. Prepare a tentative scoring key in advance of considering examinee responses.
16. Score all answers to one question before scoring the next question.
17. Make prior decisions regarding treatment of such factors as spelling and punctuation.
18. Evaluate essay responses anonymously.

Q.4:

Describe the advantages for the essay writing test?

Answer:

1. Effective for assessing higher order abilities: *analyses synthesize and evaluate*.
2. It is comparatively *less time consuming* to develop such items.
3. Emphasizes essential *communicational skills*.
4. *Guessing* is eliminated

Q.5:

Explain Item difficulty in IRT.

Answer:

Item difficulty in IRT:

The application of Item difficulty in IRT is defined as the ability at which the probability of success on the item is .**5 on a logit scale, which is also known as threshold difficulty** (see Fig. 3). An item's level of difficulty is another factor affecting an individual's probability of responding in a particular way. An item that has a high level of difficulty will be less likely to be answered correctly by an examinee with low ability than an item that has a low level of difficulty (i.e., an easy item).

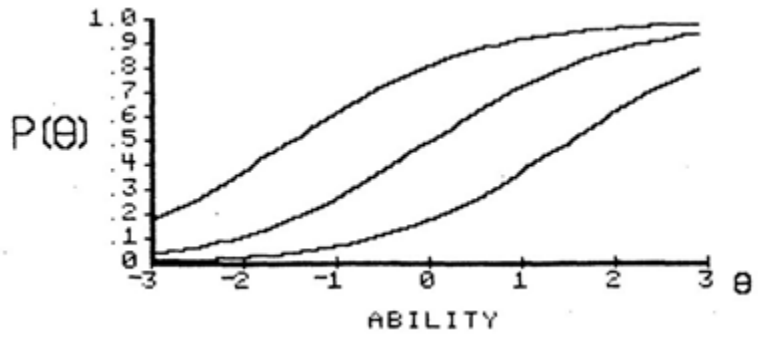


Figure 3: Three item characteristic curves with the same discrimination but different levels of difficulty (Baker, 2001, p. 10)

In Figure 2, three item characteristic curves are presented on the same graph. All have the same level of discrimination but differ with respect to difficulty. The left- hand curve represents an easy item because the probability of correct response is high for low-ability examinees and approaches 1 for high-ability examinees. The centre curve represents an item of medium difficulty because the probability of correct response is low at the lowest ability levels, around 0.5 in the middle of the ability scale and near 1 at the highest ability levels. The right-hand curve represents a hard item. The probability of correct response is low for most of the ability scale and increases only when the higher ability levels are reached. Even at the highest ability level shown (+3), the probability of correct response is only 0.8 for the most difficult item.

Q.6:

What is the Rubric, Write the element of the rubric?

Answer:

Rubric:

A rubric is an explicit set of criteria used for assessing a particular type of work or performance and provides more details than a single grade or mark. It is a set of scoring guidelines for evaluating student work.

Scores or levels identified in a scoring rubric must be descriptive, not merely judgmental in nature. **Example:** Define the level of rubric as "Writing is clear and thoughts are complete" as compared to "Excellent."

Elements of Rubric:

A rubric includes:

1. Score
2. Criteria
3. Levels of performance
4. Descriptors

Q.7:

What is the self-assessment?

Answer:

Self-assessment:

Self-assessment is a process of formative assessment during which students reflect on and evaluate the quality of their work and their learning, judge the

degree to which they reflect explicitly stated goals or criteria, identify strengths and weaknesses in their work, and revise accordingly (2007, p.160).

Q.8:

What is Analytic rubric?

Answer:

Each criterion is assessed separately by using different descriptive ratings.

Each criterion is given a separate score.

Final score is made up of adding each component parts.

It takes more time to score but gives detailed more feedback.

Q.9:

Describe the Advantages of Self-Assessment.

Answer:

Advantages:

- Self-assessment can promote learner responsibility and independence.
- Self-assessment tasks encourage student ownership of the learning.
- Self-assessment tasks shift the focus from something imposed by someone else to a potential partnership.
- Self-assessment emphasizes the formative aspects of assessment.

- Self-assessment encourages a focus on process.
- Self-assessment can accommodate diversity of learners‟ readiness, experience and backgrounds.

Q.10:

Why use peer assessment?

Answer:

- Falchikov (2007) reminds us that peer learning builds on a process that is part of our development from the earliest years of life (it is the practice of formal education and the centrality of the teacher that makes us lose sight of this).
- Peer feedback can encourage collaborative learning through interchange about what constitutes good work.
- If the course wants to promote peer learning and collaboration in other ways, then the assessment tasks need to align with this. It is also important to recognize the extra work that peer learning activities may require from students through the assessment. Boud, Cohen & Sampson (1999) observe that "if students are expected to put more effort into a course through their engagement in peer learning activities, then it may be necessary to have this effort recognized through a commensurate shift in assessment focus" (p.416).
- Peer learning draws on the "cognitive apprenticeship model" (Kvale, 2006).
- Students can help each other to make sense of the gaps in their learning and understanding and to get a more sophisticated grasp of the learning process.

- The conversation around the assessment process is enhanced. Research evidence indicates that peer feedback can be used very effectively in the development of students‟ writing skills.
- Students engaged in commentary on the work of others can heighten their own capacity for judgment and making intellectual choices.
- Students receiving feedback from their peers can get a wider range of ideas about their work to promote development and improvement.
- Peer evaluation helps to lessen the power imbalance between teachers and students and can enhance the students‟ status in the learning process.
- The focus of peer feedback can be on process, encouraging students to clarify, review and edit their ideas.
- It is possible to give immediate feedback, so formative learning can be enhanced. Peer assessment processes can help students learn how to receive and give feedback which is an important part of most work contexts.
- Peer assessment aligns with the notion that an important part of the learning process is gradually understanding and articulating the values and standards of a "community of practice" (Wenger, 1999, cited in Falchikov, 2007, p.129). Drawing on Wenger‟s ideas, Falchikov suggests that "learning involves active participation in a „community of practice‟ in which members of the community determine and structure their own practices, and construct identities in relation to these communities" (2007, p.129). Peer commentary in the assessment process initiates into the community to hear, experiment with and gradually internalize the norms of the community

Q.11:

What is the Syllabus criteria for the higher education in Pakistan?

Answer:

Q.12:

What are the objective test item types?

Answer:

There are three types of items which are included in the category of selection type items or the objective test items types

1. : multiple choice items,
2. alternative form or
3. true-false and matching.

Q.13:

Define Reliability with examples.

Answer:

Reliability refers to the extent to which assessment results are consistent. The assessment that maintains the consistency in results is known as reliable.

"Reliability is the extent to which a measurement instrument or procedure yields the same results as on repeated trails (Carmines and Zeller, 1979)"

For example, a calculator cannot be reliable if it gives wrong calculations or results. The weight of an object is not reliable if machine is not working properly. Same is the case of the results obtained from assessments.

What is meant by the term of unreliable results? The unreliable results of assessment mean inconsistency. Keeping these views in mind, we can trust the results if these are consistent. It is the evidence ofan accurate system of assessment. Keeping in view the significance of reliability, different methods are used to attain reliability. For example, test-retest procedure, the alternative–test form procedure and the split halves procedure.

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Q.1:

Define the evaluation?

Answer:

Evaluation is process of making a value judgment against intended learning outcomes and behavior, to decide quality and extent of learning.

Q.2:

What is the item difficulty?

Answer:

Item difficulty is the is a measure of the proportion of examinees who responded to an item correctly.

Q.3:

What are the importance of the student self-assessment?

Answer:

1. **Clarifying, sharing, and understanding learning intentions and criteria for success**– getting the students to really understand what their classroom experience will be and how their success will be measured.
2. **Engineering effective classroom discussions, activities, and learning tasks that elicit evidence of learning**– developing effective classroom instructional strategies that allow for the measurement of success.
3. **Providing feedback that moves learning forward**– working with students to provide them the information they need to better understand problems and solutions.
4. **Activating learners as instructional resources for one another**– getting students involved with each other in discussions and working groups can help improve student learning.
5. **Activating learners as owners of their learning**– self-regulation of learning leads to student performance improvement

Q.4:

What is the alternative form of test?

Answer:

Alternative form of question requires students to select any one of the two given categories. The categories may be True-false, yes-no, correct-incorrect or fact-opinion (Airasian, 1994). Alternative form of items is most suitable for measuring lower level learning outcomes.

Q.5:

What is the maximum and Typical performance?

Answer:

Maximum Performance:

What a person can do. For example how you drive on a drivers test. Happens when someone knows they are being observed.

Typical Performance:

what a person *will* do – ie. how you drive normally. To observe typical performance, you have to do it when people don't know they are being observed or be there on enough occasions that people get used to the examinee, or don't know what they are being tested on.

Q.6:

What is the restricted response item?

Answer:

Pose a specific problem for which student needs to recall suitable information, organize it, derive a defensible conclusion, and express it within the given limits the questions (like 400 word, 30 min. etc.).

Q.7:

What is face validity and content validity?

Answer:

Face Validity:

- Does the test appear to test what it is supposed to be tested?
- A drawing test must be related to figures, so the question paper should appear with figures. A test that asks for filling some blanks must have blanks on paper. A multiple choice question must have options along with it.

Content Validity:

Content validity refers to the contents a test is suppose to measure.

For instance, a book has ten chapters the chapters are the contents of the book.

Q.8:

Describe the curriculum in Pakistan at higher level education?

Answer:

- In 1973's constitution, Government placed education on the concurrent list.
- Federal Government took the responsibility to determine the curriculum text boards, policy, planning and standards of education.
- In December, 1976 Federal Government appointed University Grant commission (now HEC) as the competent authority to look after the curriculum revision work at bachelor level and onwards
- It also look after the degrees, certificates and diplomas awarded by degree colleges, universities and other institutions of HEC.

Q.9:

Describe the item difficulty in CTT?

Answer:

At the item level, the Classical Test Theory (CTT) model is relatively easy. CTT does not raise a complex theoretical model to relate an examinee's ability to the probability of success on a particular item. Instead, CTT collectively considers a pool of examinees and empirically examines their success rate on an item (assuming it is dichotomously scored). This success rate of a particular pool of examinees on an item, well known as the p value of the item, is used as the index for the item difficulty (Linden& Hambleton, 2004). (Actually, it is an inverse indicator of item difficulty, with higher value indicating an easier item). In CTT, the item difficulty index p (p value), is the proportion of examinees correct on an item, expresses item difficulty on an Item.

Item difficulty in CTT is simply calculated by the percentage of students that correctly answered the item as refers to the p value which range from .00 to 1.00. The values closer to 1 more easy will be the item and conversely the values near to .00 the more difficult will be the item. The values lie somewhere in the middle i.e. 0.4 to 0.6 will refer to moderate item difficulty index.

Item difficulty: the percentage of students that correctly answered the item. The range is from 0% to 100%, or more typically written as a proportion of 0.0 to 1.00. The higher the value, the easier will be the item.

Calculation: Divide the number of students who got an item correct by the total number of students who answered it.

Ideal value: Slightly higher than midway between chance (1.00 divided by the number of choices) and a perfect score (1.00) for the item. For example, on a four- alternative, multiple-choice item, the random guessing level is 1.00/4 = 0.25; therefore, the optimal difficulty level is .25 + (1.00 - .25) / 2 = 0.62. On a true-false question, the guessing level is (1.00/2 = .50) and, therefore, the optimal difficulty level is .50+(1.00- .50)/2 = 0.75. P-values above 0.90 are very easy items and should be carefully reviewed based on the instructor's purpose. For example, if the instructor is using easy "warm-up" questions or aiming for student mastery, than some items with p values above .90 may be warranted. In contrast, if an instructor is mainly interested in differences among students, these items may not be worth testing. P-values below 0.20 are very difficult items and should be reviewed for possible confusing language, removed from subsequent exams, and/or identified as an area for re-instruction. If almost all of the students get the item wrong, there is either a problem with the item or students were not able to learn the concept. However, if an instructor is trying to determine the top percentage of students that

learned a certain concept, this highly difficult item may be necessary. The optimal item difficulty depends on the question-type and on the number of possible distractors. Many test experts believe that for a maximum discrimination between high and low achievers, the optimal levels (adjusting for guessing) are:

Q.10:

Explain the charactertics of the Curve.

Answer:

The item characteristic curve is considered as the basic building block of item response theory; all the other constructs of the theory depend upon this curve. Therefore, significant attention is devoted to this curve and its role within the theory. There are two methodological properties of an item characteristic curve that are used to describe it: one is the difficulty which under item response theory describes the item functions along the ability scale. For example an easy item functions among the low- ability examinees and a hard item functions among the high-ability examinees; thus, difficulty is a location index. The second technical property is discrimination, which describes how well an item can differentiate between examinees having abilities below the item location and those having abilities above the item location (Baker, 2001).

An item characteristic curve is the graphical representation of the probability of answering an item correctly with the level of ability on the construct being measured. It gives you a picture of:

a. The item difficulty

b. Discrimination power

c. The probability of answering correctly by guessing

Q.11:

What are the steps for packing test material?

Answer:

Topic 103 (Repeat)

Q.12:

What is the Prohibited Professional Test Security Practices?

Answer:

1. Give students access to test content or secure test questions before test administration.
2. Copy reproduces or uses in any manner inconsistent with test security measures of secure assessment material.
3. Review actual test items before, during or after the test administration.
4. Fail to follow directions for the distribution and return of secure material or fail to account for any secure materials before, during or after test administration.
5. Leave a testing room unsupervised at any time.
6. Make test answers available to students or permit the use of any supplemental or reference materials during test.
7. Assist a student directly or indirectly (gestures, pointing, prompting etc.,)
8. Disclose or discuss the content of the tests with students, teachers, parents, other educators or community members before, during or after testing.
9. Making any changes to student's responses.

The End …..

Best of Luck

Remember me in your prayers..

By :

Nadia Khan ..!